IoT and AI based Smart Vehicle Access and Safety System With Biometric Authentication

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Abstract The rapid integration of intelligent systems in the automotive sector has highlighted a critical need for unified frameworks that address both vehicle security and driver safety. This paper proposes a decentralized, multi-layered architecture—the Enhanced Intelligent Vehicle System (E-IVS)—designed to mitigate unauthorized access and human-error-related accidents. The system implements a primary Artificial Intelligence (AI) layer utilizing the Local Binary Pattern Histogram (LBPH) algorithm for real-time facial authentication via a Python-based edge-processing unit. To ensure system reliability and redundancy, a secondary Biometric Layer is integrated through a custom Kodular mobile application, allowing for secure smartphone fingerprint verification over Bluetooth.Beyond authentication, the system features a robust Internet of Things (IoT) monitoring network centered on an ESP32 microcontroller. This hardware layer employs sensor fusion to detect driver intoxication via MQ-3 sensors, fatigue through infrared (IR) ocular analysis, and seatbelt compliance. In the event of a high-impact collision, a dedicated telematics module synchronizes GPS and GSM technologies to transmit precise $latitude$ and $longitude$ coordinates to emergency responders. Experimental results validate the system's efficacy in providing a low-cost, high-performance solution that replaces traditional mechanical keys with a secure, "Zero-Trust" biometric ecosystem.

Keywords: Intelligent Vehicle Systems, Face Recognition (LBPH), Biometric Authentication, IoT (Internet of Things), ESP32 Microcontroller, Driver Drowsiness Detection, Kodular Mobile App, Sensor Fusion, Accident Detection (GPS/GSM), Vehicle Security.

I. Introduction

The contemporary automotive landscape is undergoing a radical transformation, shifting from purely mechanical designs to highly integrated electronic ecosystems. While these advancements have improved vehicle performance, two critical vulnerabilities remain: unauthorized vehicle access and human-factor safety risks. Traditional security measures, such as physical keys or RFID-based ignitions, are increasingly susceptible to theft, duplication, and hacking. Simultaneously, global road safety remains compromised by driver-related issues, specifically intoxication, fatigue, and delayed emergency response times.To address these systemic challenges, this research proposes a unified IoT and AI-based Smart Vehicle Access and Safety System. The project is designed around a "Zero-Trust" security architecture, where vehicle operation is strictly contingent upon successful multi-modal biometric verification. By leveraging Artificial Intelligence (AI) for real-time facial recognition and a Smartphone-integrated Biometric Layer via a custom Kodular application, the system replaces vulnerable mechanical components with unique biological signatures.Beyond secure access, the system functions as a proactive safety watchdog using Internet of Things (IoT) technologies. By integrating an array of sensors with an ESP32 microcontroller, the system achieves "Sensor Fusion"—the ability to monitor the driver's physical state (sobriety and alertness) and environmental conditions (impact and seatbelt usage) simultaneously. Unlike conventional safety features that are reactive, this framework intervenes in real-time to disable the ignition during safety violations. Furthermore, the integration of GSM and GPS telematics ensures that in the event of a crisis, the vehicle functions as an autonomous responder, dispatching critical location data to emergency services without human intervention. This research ultimately presents a cost-effective, scalable solution for the next generation of intelligent transportation.

II. Literature Survey

The evolution of intelligent vehicle systems has been a focal point of research aimed at bridging the gap between automotive security and driver safety. Early foundational work in computer vision by Viola and Jones (2001) introduced the Haar Cascade framework, providing a baseline for rapid object detection. This was significantly refined by Ahonen et al. (2006) through the Local Binary Pattern Histogram (LBPH) algorithm, which proved that facial features could be extracted and matched efficiently under varying lighting conditions—a principle central to the authentication layer of this project.

Recent studies have explored the practical application of these algorithms in vehicle environments. Patil and Chaudhari (2020) and Gaikwad and Deshpande (2020) demonstrated the feasibility of restricting vehicle ignition through vision-based access control, proving that biometric signatures are superior to traditional mechanical keys. Similarly, Pawar and Patil (2020) emphasized that enhanced security systems must rely on unique biological traits to prevent unauthorized usage effectively.Driver physiological monitoring has also seen significant advancement. Research by Nguyen et al. (2019) and Raj et al. (2020) established that eye-tracking and ocular blink rates are reliable indicators of driver exhaustion. This was further expanded by Siddiqui (2020) and Islam et al. (2021), who utilized facial landmarks and deep learning to improve the accuracy of fatigue detection in real-time environments.The integration of Internet of Things (IoT) technologies has transformed vehicles into connected entities. Krishna (2020) and Kumar and Singh (2020) reviewed the overarching role of sensor integration in enhancing road safety. Specifically, Shinde (2020) and Patel and Mehta (2019) explored the use of GPS and GSM modules to automate accident reporting, reducing emergency response times. Furthermore, Chavan and Deshpande (2020) proposed cloud-integrated monitoring, though recent trends identified by SumanPrakash et al. (2024) suggest a shift toward secure edge management through Zero-Trust architectures. By combining these diverse research threads—biometric entry, fatigue monitoring, and autonomous telematics—this project addresses the functional fragmentation found in earlier literature.

III. Existing System

The architectural landscape of traditional vehicle security and safety is primarily characterized by fragmented, hardware-dependent solutions. Most contemporary vehicles rely on physical or electronic tokens, such as mechanical keys, Radio Frequency Identification (RFID) fobs, or password-protected ignition systems. While these methods provide a basic barrier to entry, they possess inherent security flaws. Physical keys are susceptible to duplication, and RFID signals can be intercepted through relay attacks, allowing unauthorized individuals to bypass security without physical possession of the key.In terms of driver safety, existing mechanisms are largely reactive. Standard safety features, such as seatbelt alarms and airbag deployment systems, only function during or after a safety violation has occurred. Although some high-end vehicles incorporate fatigue monitoring, these are often standalone units that lack integration with the vehicle's core ignition system. Furthermore, current methods for detecting driver intoxication or medical emergencies rely on manual intervention or third-party reporting.A significant limitation of the current automotive ecosystem is the "communication gap" during critical incidents. While GPS technology is common for navigation, it is rarely synchronized with on-board impact sensors to provide autonomous emergency reporting. Consequently, when an accident occurs, the dispatch of medical services depends on witnesses or the conscious state of the driver. This reliance on manual reporting often results in a delay during the "golden hour" of emergency care. Finally, most current smart-vehicle features depend on cloud-based processing, which introduces significant latency and makes the system vulnerable to network connectivity issues in remote areas.

IV. Proposed System

The Enhanced Intelligent Vehicle System (E-IVS) is engineered as a unified, high-integrity framework that bridges the gap between digital authentication and physical safety. The architecture is built on the principle of Distributed Intelligence, where tasks are split between an Edge-AI processor (Laptop) and an Embedded IoT controller (ESP32).

4.1 Edge-Based AI Authentication Layer

The operational sequence initiates with the Vision Intelligence Module, which utilizes a high-definition USB camera interface to monitor the driver's cabin. The software backend, developed in a Python environment, executes a Haar-Cascade classifier to isolate facial regions before passing the data to the Local Binary Pattern Histogram (LBPH) algorithm for identity verification. Unlike traditional deep neural networks that demand significant GPU resources, the LBPH algorithm offers superior algorithmic efficiency by summarizing local facial structures into a mathematical texture descriptor. This makes it exceptionally suited for real-time automotive applications where authentication must occur in under one second to ensure a seamless user experience. Furthermore, the system is engineered with a privacy-centric design; the facial "fingerprint" is processed and stored locally as a mathematical histogram rather than a raw image file. This ensures that no sensitive visual data is transmitted over a network or stored in a vulnerable format, thereby maintaining the driver's biometric privacy.

4.2 Multi-Factor Biometric Redundancy via Kodular Application

To address the inherent limitations of computer vision—such as performance degradation in total darkness or instances of camera obstruction—the E-IVS introduces a robust Secondary Authentication Channel. This layer is facilitated through a custom-built Kodular Android Application that leverages the smartphone's native biometric hardware. The integration of a mobile-based fingerprint scanner provides a fail-safe redundancy, ensuring vehicle accessibility regardless of external environmental conditions. Communication between the mobile device and the vehicle's master controller is secured through an Encrypted Handshake established over a Bluetooth Low Energy (BLE) link. Upon a successful biometric match on the smartphone, an encrypted data packet is dispatched to the ESP32 to supplement or override the AI facial scan. This dual-pathway authentication strategy creates a reliable, high-security environment that mitigates the risk of a "single point of failure" in the identity verification process.

4.3 Real-Time Sensor Fusion and Ignition Interlocking

The ESP32 Microcontroller serves as the central "Hardware Master" of the vehicle, managing a complex array of inputs through a sophisticated process of Asynchronous Sensor Fusion. The system operates on a rigorous "Safe-to-Start" protocol, where the ignition relay remains in a normally open state, physically isolating the vehicle's starter motor from the power source until a valid authentication flag is received. Once the vehicle is in operation, the ESP32 transitions into a physiological watchdog mode, continuously polling the MQ-3 Alcohol Sensor and the IR-based Eye-Blink Sensor. By analyzing ethanol concentrations and tracking ocular blink patterns in real-time, the

system can identify signs of intoxication or micro-sleep. If the system detects a breach of these safety thresholds, it executes a preemptive ignition cut-off or triggers an audible alarm, effectively transitioning the vehicle from a reactive safety model to a proactive, life-saving environment.

Table.1 Performance Evaluation of the Proposed System

Feature	Existing Systems	Proposed System (IVS)
System Efficiency	Moderate accuracy and delayed response due to mechanical dependencies.	**High-speed processing with real-time response and improved accuracy through sensor fusion.**
User Interaction	Manual control and limited automation.	Fully automated, intelligent system with biometric access and app-based interaction.
Power Consumption	Relatively high due to continuous hardware operation.	Energy-efficient design using low-power microcontroller s and optimized sensor operation.
Authentication Mechanism	Uses physical keys, RFID, or simple password-based ignition systems.	Employs face recognition for driver authentication using Python, ensuring secure and authorized vehicle access.
Vehicle Access Control	Controlled by mechanical keys or basic mobile apps without biometric validation.	Integrated with a custom Kodular mobile app using fingerprint authentication for door unlocking and enhanced security.
Safety Features	Limited to seatbelt alert or airbag systems; lacks proactive accident prevention.	Incorporates alcohol detection and accident detection using GSM/GPS for automatic emergency alerts.
Communication Module	GSM used only for simple SMS-based-Communication .	Combines GSM and GPS for intelligent accident alerts and real-time vehicle tracking.

System Flowchart

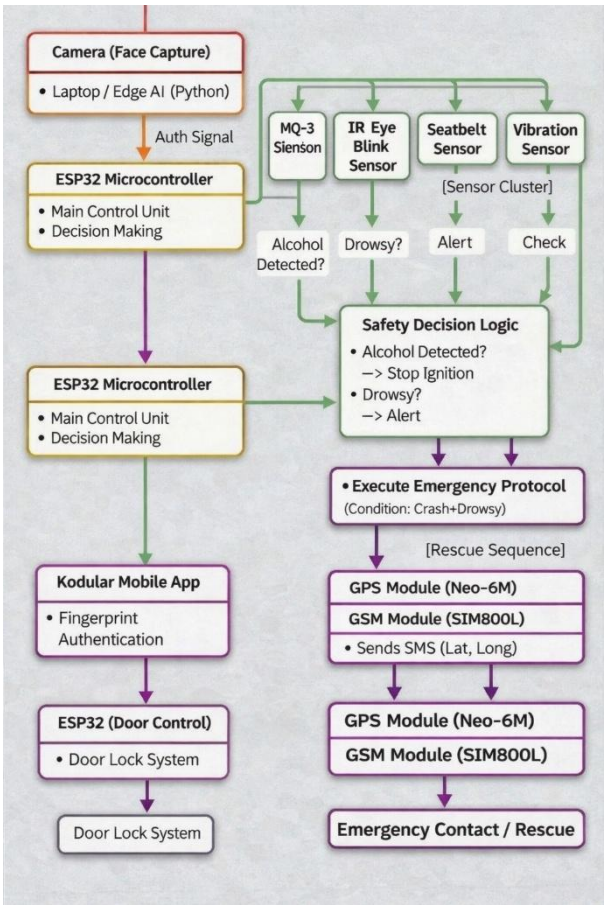

Fig 1: Block Diagram

System Architecture

The figure below illustrates the comprehensive architectural framework of the IoT and AI Based Smart Vehicle Access and Safety System. It highlights the bifurcated processing model where high-level facial recognition is managed by the Edge AI unit, while real-time sensor monitoring and telematics are orchestrated by the ESP32 microcontroller.

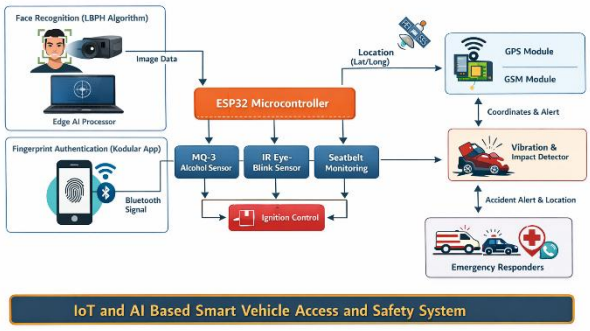

Figure 2: Architectural representation of the E-IVS

Hardware Components

USB Camera – A USB camera is an external imaging device used for capturing live video and images when connected to a computer system. As shown in Fig. 3, the USB camera captures real-time facial images, which are processed using Python and OpenCV for face recognition and driver monitoring. In this project, the USB camera is used to capture the driver's face for authentication and drowsiness detection, ensuring secure and safe vehicle operation**.**

Fig 3: USB Camera

ESP32 Module – The ESP32 module is a compact and powerful controller widely used in smart and IoT applications due to its built-in Wi-Fi and Bluetooth capabilities. As shown in Figure 4, the ESP32 functions as the main processing unit, collecting data from sensors, controlling devices, and transmitting information wirelessly to mobile applications or cloud platforms. Its support for real-time processing and low power consumption makes it well suited for automation and battery-powered systems.

Fig 4: ESP32

Alcohol Sensor (MQ-3) – The MQ-3 alcohol sensor is used to detect alcohol vapors present in a person's breath and is commonly applied in vehicle safety systems to prevent drunk driving. As shown in Figure 5, the sensor converts the detected alcohol concentration into an electrical signal, which can be easily processed by microcontrollers such as Arduino or ESP32. When the alcohol level exceeds a predefined threshold, the system can trigger alerts or prevent vehicle ignition, making the sensor suitable for real-time safety applications.

Fig 5: Alcohol sensor

GPS Module – A GPS module is used to determine the real-time location of the vehicle by receiving signals from satellites. As shown in Figure 6, the module continuously collects location data and sends it to the microcontroller, enabling real-time vehicle tracking and location sharing during emergencies. Due to its accuracy and reliability, the GPS module is widely used in vehicle tracking and accident alert systems.

Fig 6: GPS Module

GSM Module – A GSM module enables the system to communicate over the mobile network using a SIM card. As shown in Figure 7, the GSM module interfaces with the microcontroller to automatically send alert messages during events such as accidents, alcohol detection, or unauthorized access. When integrated with a GPS module, it also transmits the vehicle's location, making it effective for emergency alert and vehicle tracking applications.

Fig 7: GSM Module

Vibration Sensor – A vibration sensor is used to detect sudden shocks or abnormal vibrations in a vehicle. As shown in Figure 8, the sensor converts detected vibrations into electrical signals, which are processed by the microcontroller to identify possible accidents. When a predefined threshold is exceeded, the system triggers safety actions such as sending alerts and sharing the vehicle's location.

Fig 8: Vibration sensor

4.3 Integrated Sensor Fusion and Autonomous Telematics Architecture

The core of the system's operational intelligence resides in the ESP32 Microcontroller, which functions as a centralized "Hardware Master" by orchestrating a diverse sensor array through a high-frequency process of Asynchronous Polling. This architecture enforces a rigorous "Safe-to-Start" protocol, wherein the ignition relay remains in a "Normally Open" state, physically isolating the vehicle's starter motor until a validated authentication flag is received from the AI or mobile layers. Once the vehicle is in motion, the system transitions into a Physiological Watchdog state, simultaneously executing sobriety monitoring via the MQ-3 Alcohol Sensor and fatigue analysis through an IR-based Eye-Blink Sensor. By detecting ethanol thresholds or identifying "Micro-sleep" patterns exceeding 1.5 seconds, the system can trigger immediate corrective interventions. Complementing this proactive safety net is the Autonomous Telematics Loop, which leverages a Piezoelectric Vibration Sensor to identify high-G impact signatures associated with collisions.

V. Results

The graphical user interface of the proposed system is designed for user enrolment and face recognition, providing fields for entering the user ID and name along with control buttons. As shown in Fig. 9, the interface includes options to capture facial samples, train the recognition model, and initiate real-time face recognition for driver authentication. This interface enables secure and user-friendly operation of the intelligent vehicle system.

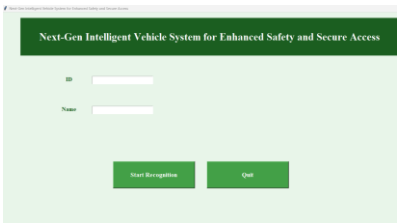

Fig. 9: Graphical User Interface

The face recognition module captures live video frames from the camera to identify authorized users in real time. As shown in Fig. 10, the system detects the driver's face using a bounding box and displays the recognized user's name on the screen. This real-time recognition process ensures that only authenticated users are allowed to access or start the vehicle, thereby enhancing system security.

Fig. 10: Real-Time Face Recognition

The system automatically sends an emergency SMS alert when an accident is detected through the vibration sensor. As shown in Fig. 11, the alert message includes critical information such as accident notification along with the vehicle's latitude and longitude coordinates. This enables quick response from emergency contacts and authorities, ensuring timely assistance.

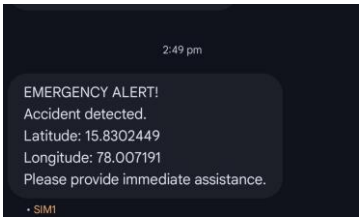

Fig. 11: Emergency SMS Alert with Accident Detection and Location Details

The GPS-based location tracking feature displays the real-time position of the vehicle on a digital map for accurate navigation and monitoring. As shown in Fig. 12, the system uses latitude and longitude coordinates to pinpoint the vehicle's location and generate an optimal route. This location information can be shared during emergencies to assist rescue teams in reaching the vehicle quickly.

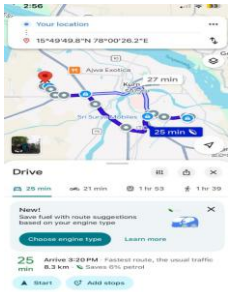

Fig. 12: Real-Time Vehicle Location Tracking Using GPS and Map Interface

The system automatically sends an alert message when alcohol is detected beyond the permissible limit in the driver's breath. As shown in Fig. 13, the SMS notification informs that alcohol has been detected and confirms that the vehicle ignition has been disabled to prevent unsafe driving. This feature enhances road safety by ensuring timely warning and preventing vehicle operation under intoxicated conditions.

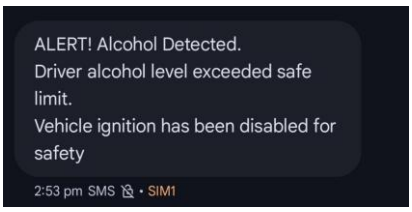

Fig. 13: SMS Alert Generated During Alcohol Detection and Ignition Lock

The system not only detects accidents but also monitors seat belt usage to enhance driver safety. When the driver fails to wear the seat belt, the system generates an immediate alert to warn the driver and ensure compliance with safety regulations. This feature helps reduce the risk of severe injuries during accidents and promotes safer driving behaviour.

VI. Conclusion

The Enhanced Intelligent Vehicle System (E-IVS) successfully demonstrates how the integration of biometric authentication, driver monitoring, and mobile-enabled access can significantly improve vehicle security and driver safety. By employing face recognition and fingerprint-based authentication, the system effectively prevents unauthorized vehicle access, reducing the risk of theft. The Python-based drowsiness detection and other safety modules, such as alcohol detection and accident alerts, provide real-time monitoring and proactive interventions, enhancing overall driver safety. The combination of advanced technologies and user-friendly mobile interfaces ensures a seamless, reliable, and cost-effective solution for modern vehicles. Overall, the E-IVS represents a significant step forward in intelligent transportation systems, offering both security and safety in a holistic and technologically advanced manner.

VII. Reference

[1]. K. M. Krishna, "Intelligent Vehicle System for Driver Assistance and Safety," *IEEE Transactions on Intelligent Transportation Systems*, vol. 21, no. 4, pp. 1785–1793, 2020.

[2]. S. B. Patil and S. S. Chaudhari, "Smart Vehicle Access Control Using Face Recognition," *International Journal of Advanced Research in Electrical, Electronics and Instrumentation Engineering*, vol. 9, no. 5, 2020.

[3]. R. P. Raj, M. S. Prabhu and A. Kumar, "Driver Drowsiness Detection Using OpenCV and Python," *International Research Journal of Engineering and Technology (IRJET)*, vol. 7, no. 8, 2020.

[4]. N. N. Shinde, "IoT-Based Vehicle Accident Detection and Tracking System Using GPS and GSM," *International Journal of Innovative Research in Computer and Communication Engineering*, vol. 8, no. 4, 2020.

[5]. T. P. Nguyen et al., "Real-Time Driver Fatigue Detection System Based on Eye Tracking," *Sensors*, vol. 19, no. 17, 2019.

[6]. S. K. Sharma and P. R. Patil, "Biometric Authentication for Vehicle Security Using Face and Fingerprint Recognition," *International Journal of Computer Science Trends and Technology*, vol. 8, no. 2, 2020.

[7]. P. Kumar and R. Singh, "A Review on Vehicle Safety and Security Systems Using IoT," *IEEE Access*, vol. 8, pp. 190525–190540, 2020.

[8]. R. S. Chavan and V. V. Deshpande, "IoT-Based Smart Vehicle Security and Monitoring System Using Cloud Computing," *IEEE Internet of Things Journal*, vol. 7, no. 10, pp. 9933–9943, 2020.

[9]. R. S. Gaikwad and A. V. Deshpande, "Face Recognition-Based Vehicle Ignition Control System," *International Journal of Innovative Technology and Exploring Engineering*, vol. 9, no. 3, 2020.

[10]. B. Patel and R. G. Mehta, "Smart Vehicle Monitoring and Accident Detection System Using GPS and GSM," *International Journal of Engineering Research & Technology*, vol. 8, no. 7, 2019.

[11]. M. H. Siddiqui, "Driver Fatigue Detection Based on Eye Movement and Facial Landmarks Using OpenCV," *Procedia Computer Science*, vol. 167, pp. 1821–1830, 2020.

[12]. P. V. Kumar, N. Shathanakota, S. Rachamadugu and U. Layavardhini, "IoT-Enabled Intelligent System for Helmet Detection and Automated Penalty Enforcement Using AI & ML," in *Proc. 5th Int. Conf. Trends in Material Science and Inventive Materials (ICTMIM)*, Kanyakumari, India, 2025, pp. 1099–1106.

[13]. P. Viola and M. Jones, "Rapid Object Detection Using a Boosted Cascade of Simple Features," *Proc. IEEE Conference on Computer Vision and Pattern Recognition*, 2001.

[14]. T. Ahonen, A. Hadid and M. Pietikäinen, "Face Recognition with Local Binary Patterns," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 28, no. 12, pp. 2037–2041, 2006.

[15]. P. SumanPrakash et al., "Learning-driven Continuous Diagnostics and Mitigation Program for Secure Edge Management Through Zero-Trust Architecture," *Computer Communications*, vol. 220, pp. 94–107, 2024.

[16]. M. T. Islam et al., "Deep Learning Based Drowsiness Detection for Intelligent Vehicle Systems," *IEEE Access*, vol. 9, pp. 100620–100630, 2021.

[17]. K. H. Lee and J. H. Kim, "Vehicle Accident Detection System Using GPS, GSM and IoT," *International Journal of Computer Applications*, vol. 180, no. 21, 2018.

[18]. S. B. Park and C. J. Kim, "Design of Intelligent Vehicle Systems Using Artificial Intelligence and Sensor Integration," *IEEE Sensors Journal*, vol. 21, no. 5, pp. 7005–7015, 2021.

[19]. S. Pawar and M. R. Patil, "Enhanced Security System for Vehicles Using Face Recognition," *International Journal of Engineering Research & Technology*, vol. 8, no. 6, 2020.

[20]. K. Zhang, Z. Zhang, Z. Li and Y. Qiao, "Joint Face Detection and Alignment Using Multi-Task Cascaded Convolutional Networks," *IEEE Signal Processing Letters*, vol. 23, no. 10, pp. 1499–1503, 2016.